Cyber Security

M Saeed Siddik

Assiatnt Professor

Institute of Infomration Technology (IIT),

University of Dhaka

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Understanding Social Engineering

Social engineering is the art of tricking people into giving up confidential information or performing actions that compromise security.

Instead of directly attacking computer systems, attackers use psychology, persuasion, and deception to get access to sensitive data such as passwords, financial details, or company secrets.

Understanding Social Engineering

* Source: https://content.kaspersky-labs.com

The Human Side of Cybersecurity

- Most people think cybercrime happens through complex codes, viruses, or technical hacks.
- In reality, many attacks start with human interaction, not with machines.
- Attackers realize that it’s often easier to trick a person than to break a firewall.

Example Scenaio Human Side of Cybersecurity

“Your account has been temporarily locked. Please verify your information to restore access.”

Social engineering is powerful because it targets human psychology

Human Emotion	**How Attackers Use It**
Fear	“Your account will be suspended in 24 hours.”
Greed	“You’ve won a $500 gift card!”
Curiosity	“Secret celebrity news – click to read.”
Trust	“I’m from IT support. Please share your login.”
Helpfulness	“Can you reset my password? I’m locked out.”

Phishing

- The most common type.

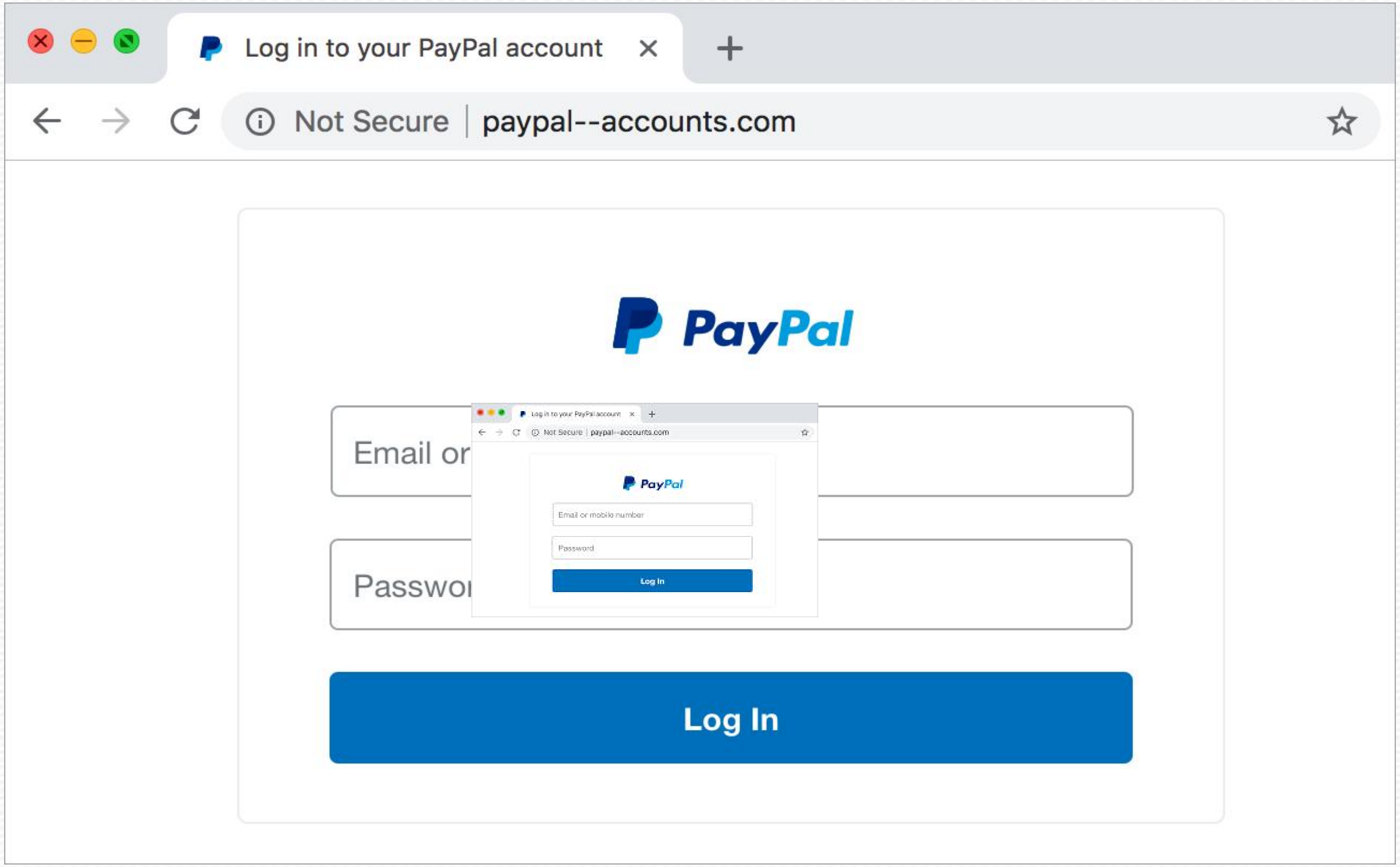
- Attackers send emails or messages pretending to be legitimate institutions.

- These messages often ask to click a link or download a file.

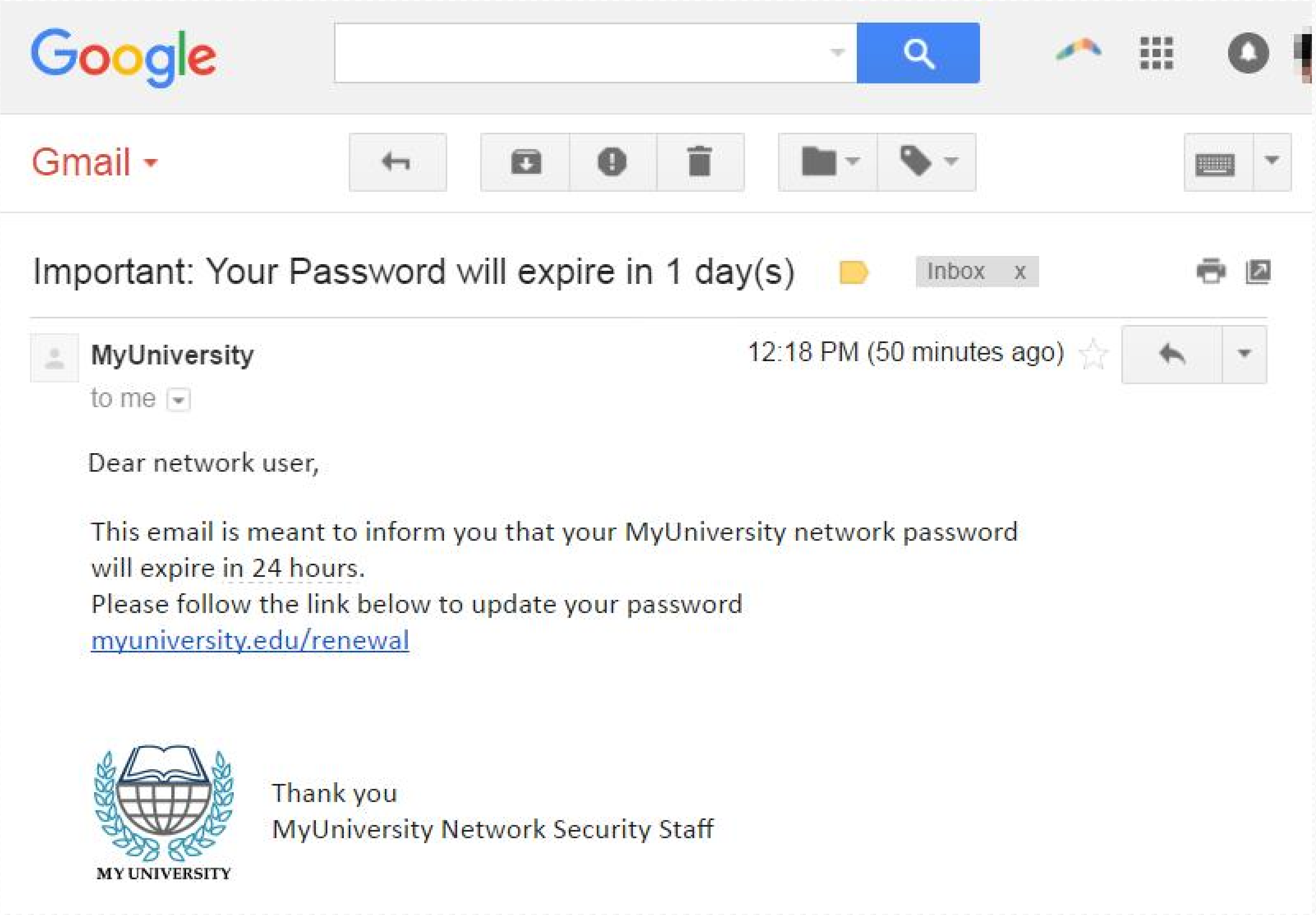
Phishing

When a hacker launches a phishing attack, **he or she is trying to trick you into believing that the message is from a legitimate source** so that you will click a link or download an attachment.

Phishing Example



Phishing Example



Phishing Example

Vishing

- Vishing, or "voice phishing," is a social engineering attack that uses phone calls or voicemails to manipulate victims into divulging information.

- Attackers use Voice over Internet Protocol (VoIP) technology to make thousands of automated calls and often use caller ID spoofing to disguise their true identity.

- Posing as a bank representative, tech support agent, or government official to gain the victim's trust.

Vishing


Scammers combine phishing emails + follow-up phone calls. The email primes the victim, while the call “verifies” the request, making the scam feel more authentic.
PHISHING EMAIL
FOLLOW-UP CALL
How to Prevent It: Train staff to recognize cross-channel manipulation. Policies should mandate callback verification before any action.
keepnet

Smishing

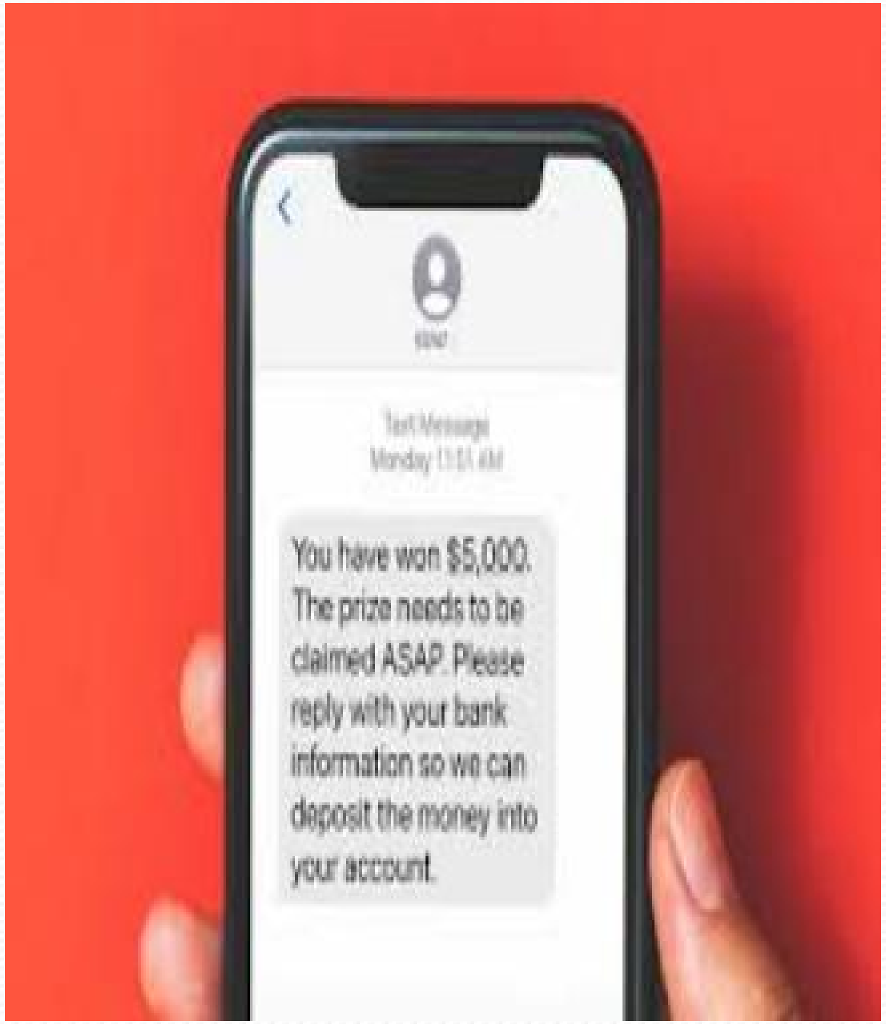
- Smishing is a social engineering attack that uses fraudulent text messages (SMS) to trick victims into revealing personal data.

- A text message, often appearing to be from a bank, delivery service, or retail giant, contains a malicious link or a number to call.

- Example: Sending a fraudulent bank fraud alert or a message about a failed delivery to prompt immediate action.

Smishing


[UPS] Tracking Number UPS306724783958ID.
Our driver can't find your address, and your package is still pending. And Our driver will redeliver tomorrow.
Please provide your complete address at https://ups-tracking-id11105.delivery/redelivery#id60959 to schedule redelivery.


You have won $5,000. The prize needs to be claimed ASAP. Please reply with your bank information so we can deposit the money into your account.

Pretexting

- Pretexting involves creating a believable, fabricated scenario (a "pretext") to gain a victim's trust and extract information

- The attacker assumes a false identity, such as an IT technician, or an auditor, to create a convincing narrative. This can occur over the phone, in person, or via email.

- Gathering background information from social media and public sources to make the story more credible.

Impersonation

- Impersonation is the act of a malicious actor pretending to be someone else to steal sensitive data. It is a key element of many social engineering attacks, including pretexting and phishing.

- An attacker creates a fake social media profile using a company's logo and name. When a customer complains on social media, the attacker replies, impersonating customer service, and directs the customer to a malicious website to "solve" the problem.

Internet Safety: Protecting Yourself Online

- refers to the knowledge and practices that help protect individuals and their data from harm or risk when using the internet.

- encompasses everything from protecting your personal information to preventing malware and avoiding online scams.

Use Private Browser

- Always use private mode or incognito mode browser in public computer.

- Delete the Cache and browesing history.

- Login the browser account and secure password manager

2-Phase Authentication (MFA)

- This adds an extra layer of security to your accounts. After entering your password, you're required to provide a second form of verification.

- This could be a code from an authenticator app like Google Authenticator, a one-time code sent to your phone via SMS, or using a physical security key.

- Even if a cybercriminal steals your password, they cannot access your account without this second factor.

Strong, Unique Passwords

- A strong password should be at least 12 characters long and a combination of uppercase letters, lowercase letters, numbers, and symbols.

- Avoid using easily guessable information like birthdays, roll number, or common words.

- Generating and securely storing complex passwords for you so you only have to remember one master password.

Phishing Awareness

- Be cautious of suspicious emails, texts, or links.

- Red flags include urgent or threatening language ("Account will be suspended!"), poor grammar and spelling, generic greetings ("Dear Customer"), and URLs that don't match the sender's real website

- Always hover over a link to see the real URL before clicking, and never download attachments from an unknown sender.

Phishing Website Check

https://checkphish.bolster.ai

https://www.bitdefender.com/en-us/consumer/link-checker

https://www.drlinkcheck.com

Secure Wi-Fi

- Public, unsecured Wi-Fi networks in places like coffee shops or airports are often not encrypted, making it easy for hackers to intercept your data.

- Avoid conducting sensitive activities like online banking, shopping, or accessing work files on these networks.

- If you must use public Wi-Fi, always connect through a Virtual Private Network (VPN) to encrypt your internet traffic and protect your privacy.

Security Best Practices

- Don’t share confidential information unless you initiated the contact.
- Verify requests through official channels.
- Enable multi-factor authentication (MFA).
- Use strong, unique passwords and a password manager.
- Regularly update software and browsers.
- Report suspicious emails or calls to your organization or service provider.

Awareness is the First Line of Defense

Checklist

- Does this message create urgency or fear?
- Is the sender’s address or phone number slightly unusual?
- Is there a request for sensitive data?
- Does the link look suspicious? Hover over it before clicking.
- Are there spelling or grammar mistakes?

Dark Web

Internet has layers

The dark web is neither inherently mystical nor uniformly criminal; it’s a set of sites that prioritize anonymity and privacy.

The dark web consists of websites that are accessible only through specific anonymizing software and protocols (for example, Tor).

Internet has layers

Surface Web – Public and searchable (Google, news, social media).

Deep Web – Private areas (emails server, databases).

Dark Web – Hidden networks requiring special tools (e.g., Tor, VPN).

How the Dark Web Works

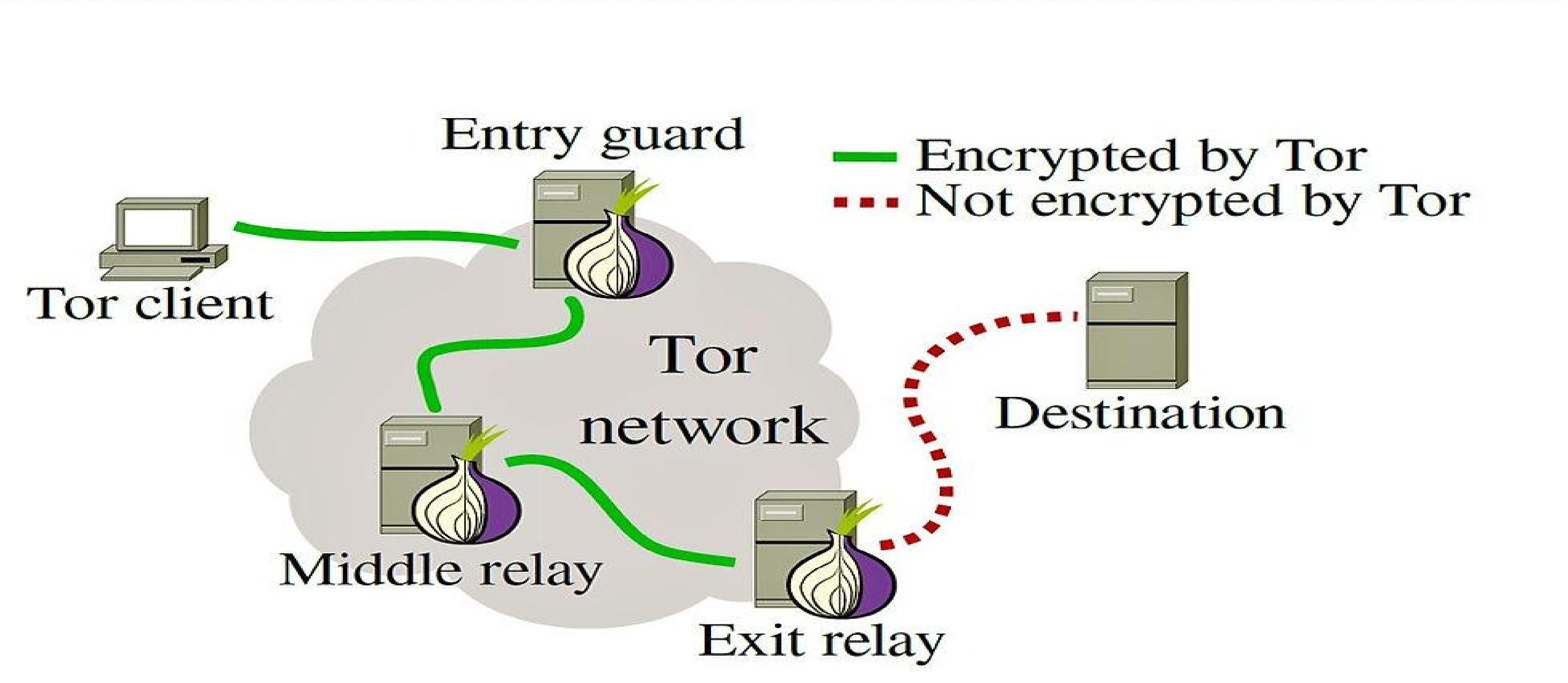
Uses Tor (The Onion Router) network to anonymize users.

Traffic is encrypted and passed through several nodes before reaching its destination.

Each node knows only its next hop, not the full path.

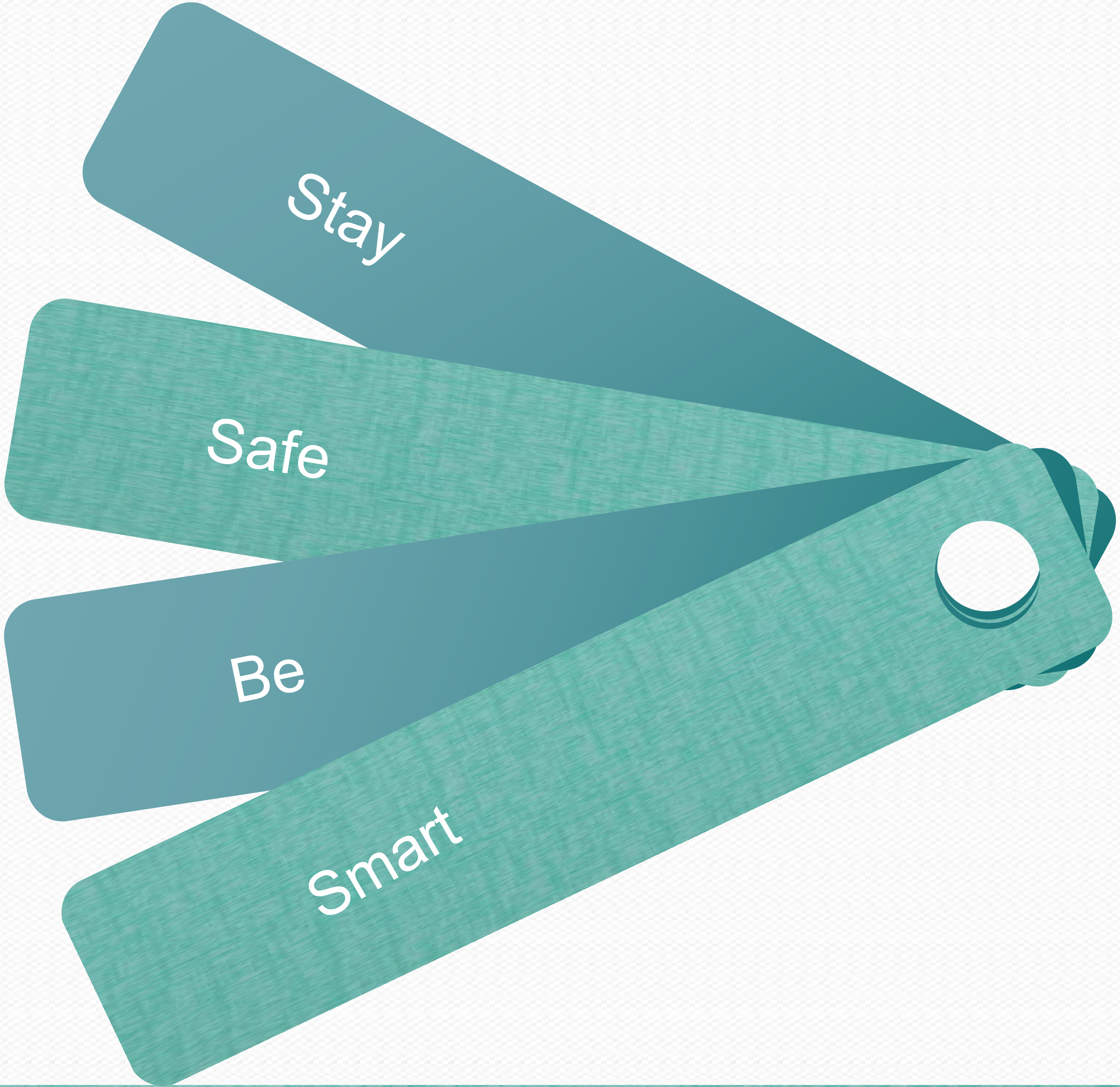
Tor hides who you are and where you connect from.

How the Dark Web Works



Access to Dark Web Works

- Verify and launch Tor safely
- Set Tor's security to a safe level
- Use an isolated environment
- Find trustworthy, legal .onion sites only
  - propublica.securedrop.tor.onion
- Never sign in with personal accounts or reuse credentials
- Avoid marketplaces, financial transactions, or file sharing
- Have a safe exit


Stay
Safe
Be
Smart

THANKS